

- 1) The angular momentum of a system remains constant
 - A) when the linear momentum and the energy are constant.
 - B) all the time since it is a conserved quantity.
 - C) when no net external force acts on the system.
 - D) when no torque acts on the system.
 - E) when the total kinetic energy is constant.
- 2) A 95 N force exerted at the end of a 0.50 m long torque wrench gives rise to a torque of 15 N · m. What is the angle (assumed to be less than 90°) between the wrench handle and the direction of the applied force?
 - A) 22°
 - B) 25°
 - C) 14°
 - D) 18°
- 3) A 72.0-kg person pushes on a small doorknob with a force of 5.00 N perpendicular to the surface of the door. The doorknob is located 0.800 m from axis of the frictionless hinges of the door. The door begins to rotate with an angular acceleration of 2.00 rad/s². What is the moment of inertia of the door about the hinges?
 - A) 2.00 kg · m²
 - B) 7.52 kg · m²
 - C) 2.74 kg · m²
 - D) 0.684 kg · m²
 - E) 4.28 kg · m²
- 4) A torque of 12 N · m is applied to a solid, uniform disk of radius 0.50 m, causing the disk to accelerate at 2.4 rad/s². What is the mass of the disk?
 - A) 10 kg
 - B) 30 kg
 - C) 40 kg
 - D) 20 kg
- 5) A uniform solid 5.25-kg cylinder is released from rest and rolls without slipping down an inclined plane inclined at 18° to the horizontal. How fast is it moving after it has rolled 2.2 m down the plane (this 2.2 m is NOT vertical height)?
 - A) 5.2 m/s
 - B) 3.0 m/s
 - C) 4.3 m/s
 - D) 2.6 m/s
 - E) 3.7 m/s
- 6) A figure skater rotating at 5.00 rad/s with arms extended has a moment of inertia of 2.25 kg · m². If the arms are pulled in so the moment of inertia decreases to 1.80 kg · m², what is the final angular speed?
 - A) 0.810 rad/s
 - B) 2.25 rad/s
 - C) 4.60 rad/s
 - D) 1.76 rad/s
 - E) 6.25 rad/s
- 7) A 5.0-m radius playground merry-go-round with a moment of inertia of 2000 kg · m² is rotating freely with an angular speed of 1.0 rad/s. Two people, each having a mass of 60 kg, are standing right outside the edge of the merry-go-round and step on it with negligible speed. What is the angular speed of the merry-go-round right after the two people have stepped on?
 - A) 0.60 rad/s
 - B) 0.80 rad/s
 - C) 0.67 rad/s
 - D) 0.20 rad/s
 - E) 0.40 rad/s

- 8) A nonuniform, 80.0-g, meterstick balances when the support is placed at the 51.0-cm mark. At what location on the meterstick should a 5.00-g tack be placed so that the stick will balance at the 50.0 cm mark?
- A) 66.0 cm B) 34.0 cm C) 16.0 cm D) 35.0 cm E) 67.0 cm
- 9) The reason an astronaut in an earth satellite feels weightless is that
- A) the astronaut is falling.
 B) this is a psychological effect associated with rapid motion.
 C) the astronaut is beyond the range of the earth's gravity.
 D) the astronaut is at a point in space where the effects of the moon's gravity and the earth's gravity cancel.
 E) the astronaut's acceleration is zero.
- 10) A meteoroid, heading straight for Earth, has a speed of 14.8 km/s relative to the center of Earth as it crosses our moon's orbit, a distance of 3.84×10^8 m from the earth's center. What is the meteoroid's speed as it hits the earth? You can neglect the effects of the moon, Earth's atmosphere, and any motion of the earth. ($G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$, $M_{\text{earth}} = 5.97 \times 10^{24} \text{ kg}$)
- A) 18.5 km/s B) 21.5 km/s C) 87.3 km/s D) 32.4 km/s
- 11) Find the orbital speed of an ice cube in the rings of Saturn. The mass of Saturn is $5.68 \times 10^{26} \text{ kg}$ and the rings have an average radius of 100,000 km. ($G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$)
- A) 27.5 km/s B) 1.95 km/s C) 13.8 km/s D) 19.5 km/s
- 12) Suppose we want a satellite to revolve around the earth 5 times a day. What should be the radius of its orbit?
- A) $0.690 \times 10^7 \text{ m}$ B) $2.11 \times 10^7 \text{ m}$ C) $1.44 \times 10^7 \text{ m}$ D) $7.22 \times 10^7 \text{ m}$
- 13) A 0.025-kg block on a horizontal frictionless surface is attached to an ideal massless spring whose spring constant is 150 N/m. The block is pulled from its equilibrium position at $x = 0.00 \text{ m}$ to a displacement $x = +0.080 \text{ m}$ and is released from rest. The block then executes simple harmonic motion along the horizontal x -axis. When the displacement is $x = 0.024 \text{ m}$, what is the kinetic energy of the block?
- A) 0.44 J B) 0.52 J C) 0.0 J D) 1.0 J E) 0.88 J
- 14) A 1.6-kg block on a horizontal frictionless surface is attached to an ideal massless spring whose spring constant is 190 N/m. The block is pulled from its equilibrium position at $x = 0.00 \text{ m}$ to a displacement $x = +0.080 \text{ m}$ and is released from rest. The block then executes simple harmonic motion along the horizontal x -axis. What is the velocity of the block at time $t = 0.40 \text{ s}$?
- A) -0.82 m/s B) 0.82 m/s C) 0.00 m/s D) -0.30 m/s E) 0.30 m/s
- 15) A uniform meter stick is freely pivoted about the 0.20-m mark. If it is allowed to swing in a vertical plane with a small amplitude and friction, what is the frequency of its oscillations?
- A) 0.66 Hz B) 0.55 Hz C) 1.1 Hz D) 1.3 Hz E) 0.92 Hz